



EXAM PAPERS PRACTICE

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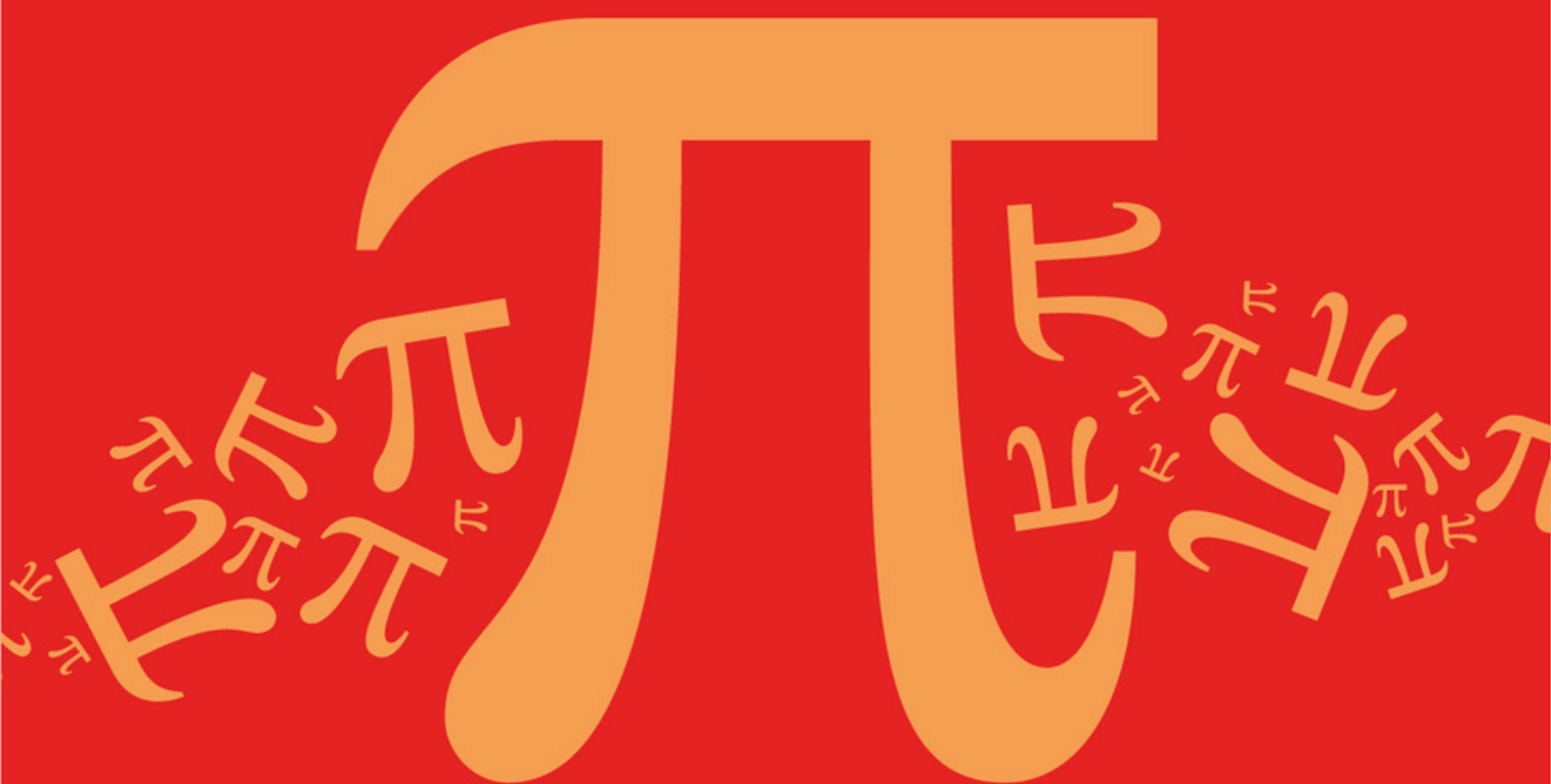
Practice questions created by actual
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Detailed mark scheme

Suitable for all boards

Designed to test your ability and
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AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B3: Complex Conjugate

Mark Scheme

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EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	10
Total 1 mark			

Q2.

Marking Instructions	AO	Marks	Typical Solution
Defines generalised z and z^* in Cartesian or polar form	AO1.2	B1	Let $z = a + bi$ then $z^* = a - bi$
Expands and simplifies zz^* and $ z ^2$ (at least one correct)	AO1.1b	M1	$zz^* - z ^2$ $= (a + bi)(a - bi) - (\sqrt{a^2 + b^2})^2$ $= a^2 + abi - abi - (bi)^2 - (a^2 + b^2)$ $= a^2 + b^2 - (a^2 + b^2)$ $= 0$ AG
Completes a well structured argument to prove the required result. AG Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	ALT Let $z = re^{i\theta}$ then $z^* = re^{-i\theta}$ $zz^* - z ^2 = re^{i\theta}re^{-i\theta} - r^2$ $= r^2e^{i\theta-i\theta} - r^2$ $= r^2 - r^2$ $= 0$ AG
Total 3 marks			

Q3.

(a) (w=)
$$\frac{-6 \pm \sqrt{36 - 4(34)}}{2} \left\{ \frac{-6 \pm \sqrt{-100}}{2} \right\}$$

Correct substitution into quadratic formula OE

M1

$$= \frac{-6 \pm 10i}{2}$$

$$\sqrt{-100} = 10i \text{ or } \sqrt{-100} / 2 = 5i$$

B1

$$= -3 \pm 5i$$

$$-3 \pm 5i \text{ (} p = -3, q = \pm 5 \text{)}$$

NMS mark as 3/3 or 0/3

A1

3

(b) (i) $z = i(1 + i)(2 + i) = i(2 + 3i + i^2) = 2i + 3i^2 + i^3$

Attempt to expand all brackets.

M1

$$= 2i + 3(-1) + i(-1)$$

$i^2 = -1$ used at least once

B1

$$= -3 + i$$

$$-3 + i \text{ (} a = -3, b = 1 \text{)}$$

A1

3

(ii) $z^* = -3 - i$

$$-3 + i + m(-3 - i) = ni$$

OE Ft c's $a - bi$

B1F

$$\Rightarrow -3 - 3m = 0; 1 - m = n$$

Equating **both** real parts and the imag. parts, PI by next line

M1

$$\Rightarrow m = -1, n = 2$$

Both correct

A1

3

[9]

Q4.

(a) (i) $(z - 2i)^* = (x + yi - 2i)^* = x + (2 - y)i$

$x + 2i - yi$ OE rearrangement

B1

1

(ii) $(z - 2i)^* = 4iz + 3 = 4ix + 4i^2y + 3 = 4ix - 4y + 3$

$$x + (2 - y)i = 4ix - 4y + 3 \quad (\#)$$

$$i^2 = -1 \text{ used}$$

B1

Real parts: $x = -4y + 3$

Imaginary parts: $2 - y = 4x$

*Attempting to equate, without mixing real and imaginary terms, **both** the real parts and the imag. parts for the c's eqn (#).*

M1

If not corrected, ft on $[c's(a)(i)] = 4ix - 4y + 3$ provided both the resulting linear equations have non zero x, y and const terms

A1F

$$y = \frac{2}{3}, \quad x = \frac{1}{3}$$

Solving correct equations, to obtain either $x = \frac{1}{3}$ OE or

$$y = \frac{2}{3} \text{ OE}$$

A1

$(z =) \frac{1}{3} + \frac{2}{3}i$
 $\frac{1}{3} + \frac{2}{3}i$

A1

5

(b) (One of the) coefficients (of the quadratic equation is) not real.

OE eg Sum of roots is $-10i$ so p cannot be real if roots are $p \pm qi$

E1

1

[7]

Q5.

(a) (i) $x = \pm 3i$

$$\pm 3i \quad (a = 0, b = \pm 3)$$

B1

1

(ii) $x = -2 \pm 3i$

If not correct, ft wrong answer(s) to (i) provided (i) has a non-zero b value

B1F

1

(b) (i) $(1 + x)^3 = 1 + 3x + 3x^2 + x^3$

Terms simplified in any order.

B1

(ii) $(1 + 2i)^3 = 1 + 3(2i) + 3(2i)^2 + (2i)^3 = 1 + 3(2i) + 3(4i^2) + (8i^3)$

Replacing x in (b)(i) by 2i, squaring and cubing correctly, only ft on c's wrong non-zero coefficients from (b)(i).

B1F

$= 1 + 3(2i) + 3(4)(-1) + (8)(-i)$

Use of $i^2 = -1$ at least once.

M1

$= -11 - 2i$

$-11 - 2i$ ($a = -11$, $b = -2$)

A1

3

(iii) $z^* - z^3 = 1 - 2i - (-11 - 2i)$

Use of $z^ = 1 - 2i$*

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M1

$= 12$

If not correct, only ft on $1 - 2i$ - c's (b)(ii) if c's (b)(ii) answer is of the form $a + bi$ with $a \neq 0$ and $b \neq 0$

A1F

2

[8]

Q6.

(a) $i(z + 7) + 3(z^* - i) = i(x + iy + 7) + 3(x - iy - i)$

M1 for use of $z^ = x - iy$*

M1

$= ix - y + 7i + 3x - 3iy - 3i$

M1 for $i^2 y = -y$

M1

$$= 3x - y + i(x - 3y + 4)$$

If the five terms correct but not grouped into Real and Imaginary parts, allow A1 retrospectively provided the correct two expressions used in the M1 line in (b)

A1

3

(b) $3x - y = 0, \quad x - 3y + 4 = 0$

Attempting to equate all Real parts to zero and all Imaginary parts to zero

M1

$$x - 9x + 4 = 0 \quad (\text{or eg } y - 9y + 12 = 0)$$

*A correct equation in either x or y
PI by correct final answer*

A1

Solving to give $z = \frac{1}{2} + \frac{3}{2}i$

Allow $x = \frac{1}{2}, y = \frac{3}{2}$

A1

3

[6]

Q7.

(a) (i) $z_1^2 = \frac{1}{4} - i + i^2 = -\frac{3}{4} = i$

M1 for use of $i^2 = -1$

M1A1

2

(ii) $\text{LHS} = -\frac{3}{4} - i + \frac{1}{2} + i + \frac{1}{4} = 0$

AG; M1 for z^ correct*

M1A1

2

(b) $\text{LHS} = -\frac{3}{4} + i + \frac{1}{2} - i + \frac{1}{4} = 0$

AG; M1 for correct

M1A1

2

(c) $z \text{ real} \Rightarrow z^* = z$

Clearly stated

M1

Discr't zero or correct factorisation
AG

A1

2

[8]

Q8.

- (a) Use of $z^* = x - iy$

M1

$$(z - i)(z^* - i) = (x^2 + y^2 - 1) - 2ix$$

A1 may be earned in (b)

m1A1

3

- (b) Equating R and I parts

M1

$$-2x = -8 \text{ so } x = 4$$

A1

$$16 + y^2 - 1 = 24 \text{ so } y = \pm 3 \text{ (} z = 4 \pm 3i \text{)}$$

A0 if $x = -4$ used

m1A1

4

[7]

Q9.

- (a) $z^2 = 1 + 2i + i^2 = 2i$

M1 for use of $i^2 = -1$

M1A1

2

- (b) $z^8 = (2i)^4$

or equivalent complete method

M1

$$\dots = 16i^4 = 16$$

convincingly shown (AG)

A1

2

(c) $(z^*)^2 = (1 - i)^2$
 for use of $z^* = 1 - i$

M1

... = $-2i = -z^2$
 convincingly shown (AG)

A1

2

[6]

Q10.

(a) Use of $z^* = x - iy$

M1

Use of $i^2 = -1$

M1

$(1 - 2i)z - z^* = 2y + i(2y - 2x)$
 A1 if one numerical error made

A2,1

4

(b) $2y = 20$, $2y - 2x = 10$
 equate and attempt to solve

M1

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so $z = 5 + 10i$
 allow $x = 5$, $y = 10$

A1

2

[6]

Q11.

(a) Other root is $2 - 3i$

B1

1

(b) Sum of roots = 4
 ft error in (a)

B1F

So $b = -4$
 ft wrong value for sum

B1F

Product is 13

B1

So $c = 13$

ft wrong value for product

B1F

Alternative:

Substituting $2 + 3i$ into equation

M1

Equating R and I parts

m1

$$12 + 3b = 0, \text{ so } b = -4$$

A1

$$-5 + 2b + c = 0, \text{ so } c = 13$$

ft wrong value for b

A1F

4

EXAM PAPERS PRACTICE [5]

Q12.

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(a) (i) $z^2 = (x^2 - 4) + i(4x)$

M1 for use of $i^2 = -1$

M1A1

R and I parts clearly indicated

Condone inclusion of i in I part

ft one numerical error

A1F

3

(ii) $z^2 + 2z^* = (x^2 + 2x - 4) + i(4x - 4)$

M1 for correct use of conjugate

ft numerical error in (i)

M1A1F

2

(b) $z^2 + 2z^*$ real if imaginary part zero

M1

... ie if $x = 1$

ft provided imaginary part linear

A1F

2

[7]

Q13.

$$z_1 + 4i z_1^* = (2 + i) + 4i(2 - i)$$

Use of conjugate

M1

$$\dots = (2 + i) + (8i + 4)$$

Use of $i^2 = -1$

M1

$$\dots = 6 + 9i, \text{ so } x = 6 \text{ and } y = 3$$

M1 for equating Real and imaginary parts

M1A1

[4]

Q14.

(a) Use of $z^* = x - iy$

M1

Use of $i^2 = -1$

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M1

$$3iz + 2z^* = (2x - 3y) + i(3x - 2y)$$

Condone inclusion of i in I part

A1

3

(b) Equating R and I parts

M1

$$2x - 3y = 7, 3x - 2y = 8$$

with attempt to solve

m1

$$z = 2 - i$$

Allow $x = 2, y = -1$

A1

3

[6]

Q15.

- (a) Use of $z^* = x - iy$

M1

$$z - 3iz^* = x + iy - 3ix - 3y$$

Condone sign error here

m1

$$R = x - 3y, I = -3x + y$$

*Condone inclusion of i in I
Allow if correct in (b)*

A1

3

- (b) $x - 3y = 16, -3x + y = 0$

M1

Elimination of x or y

m1

$$z = -2 - 6i$$

*Accept $x = -2, y = -6$;
ft $x + 3y$ for $x - 3y$*

A1F

3

[6]

Q16.

- (a) (i) Full expansion of product

M1

Use of $i^2 = -1$

m1

$$(2 + \sqrt{5}i)(\sqrt{5} - i) = 3\sqrt{5} + 3i$$

$\sqrt{5}\sqrt{5} = 5$ must be used – Accept not fully simplified

A1

3

- (ii) $z^* = x - iy (= \sqrt{5} + i)$

M1

Hence result

Convincingly shown (AG)

A1

2

- (b) (i) Other root is $\sqrt{5} + i$

B1

1

- (ii) Sum of roots is $2\sqrt{5}$

B1

Product is 6

M1A1

3

- (iii) $p = -2\sqrt{5}, q = 6$

B1

ft wrong answers in (ii)

B1ft

2

[11]

EXAM PAPERS PRACTICE

Q17.

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- (a) $(z + i)^* = x - iy - i$

B2

2

- (b) $\dots = 2ix - 2y + 1$
 $i^2 = -1$ used at some stage

M1

Equating R and I parts

involving at least 5 terms in all

M1

$$x = -2y + 1, -y - 1 = 2x$$

ft one sign error in (a)

A1ft

$$z = -1 + i$$

ditto; allow $x = -1, y = 1$

m1A1ft

5

[7]

Q18.

(a) $z^* = x - iy$

B1

1

(b) $R = 2x - y$

$i^2 = -1$ *must be used*

$I = -x + 2y$

*Condone $I = i(x + 2y)$;
Answers may appear in (c)*

B1

B1

2

(c) Equating R and/or I parts

M1

Attempt to solve sim equations

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$z = 1 + 2i$

Allow $x = 1, y = 2$

A1

3

[6]

Examiner reports

Q1.

This question was answered very well, with almost all students selecting the correct answer of 10.

Q3.

Many correct answers were submitted for parts (a) and (b)(i) in this question which tested the complex numbers section of the specification. In part (b)(ii) a common mistake was an incorrect z^* value either $i + 3$ following $z = i - 3$ in (b)(i) or, more frequently, z^* written incorrectly as $i(1 - i)(2 - i)$. Students who obtained an incorrect value for z in part (b)(i) were awarded credit for a correct follow through value for z^* in part (b)(ii). Most students showed that they appreciated the need to equate the real parts and the imaginary parts for their resulting equation.

Q4.

This question which tested the complex numbers section of the specification proved, as expected, to be more demanding than the corresponding questions in recent MFP1 papers. In part (a)(i) a common wrong answer for $(z - 2i)^*$ was $x + iy + 2i$. Those students who started by writing $(x + iy - 2i)^*$ were generally more successful. There were some very good solutions to part (a)(ii), but some able students, having found the correct values for x and y , either stopped at that point or else gave their final answer as $\frac{1}{3} + \frac{4}{3}i$ which indicated that they thought that solving the given equation meant finding the value of $(z - 2i)^*$ and not the value of z . It was quite common to see some students making no attempt to equate real parts and imaginary parts, with final answers left in a form similar to $x + 4y - 3 + (2 - y - 4x)i$. The final part of the question was answered well by those who based their statement on:

- coefficients must be real for conjugate roots or
- sum of roots ($= 2p$) cannot be $-10i$ if p is real or
- product of the roots ($p^2 + q^2$) cannot be -29 if p and q are real.

However, correct statements were in the minority. Common wrong answers for example, 'it is not the difference of two squares' suggested that students were treating the roots $p \pm iq$ as the factors of the quadratic expression $z^2 + 10iz - 29$.

Q5.

In this question on complex numbers, part (a) generally caused more problems than part (b).

In part (a)(i) a significant number of candidates failed to include the $\pm$, but, as follow through was accepted, most of these scored the mark in part (a)(ii).

Almost all candidates correctly expanded $(1 + x)^3$ in part (b)(i) and used their expansion within part (b)(ii) but there were many examples of incorrect / no squaring and cubing of the 2. The meaning and use of the complex conjugate z^* was well understood and a majority of candidates went on to give a correct follow through answer in part (b)(iii).

Q6.

This question which tested complex numbers was generally answered very well with the majority of students scoring full marks. In part (a) almost all students correctly used $z^* = x - iy$ and $i^2 = -1$ and errors were generally restricted to either incorrect expansion of brackets or incorrect collecting of like terms. Part (b) caused slightly more problems with some students failing to even equate real parts and imaginary parts separately to zero.

Q7.

The first six marks in this question were obtained with apparent ease in the great majority of scripts. Part (c) usually produced no further marks as the candidates omitted the star from z^* without explaining why this was legitimate. Many candidates seemed to treat the given equation as a quadratic, despite the fact that two roots had already been found and two more were now being asked for.

Q8.

Part (a) of this question was a minefield for some candidates, who showed the terms of their expansions very untidily and made errors, very often sign errors, in reaching a simplified form. In part (b), the technique of equating real and imaginary parts was well known and, in most cases, the two correct values were obtained for the imaginary part of z .

Q9.

Full marks were usually awarded in this question. Answers to part (b) were sometimes very laborious but eventually correct, but by contrast some answers were so brief as to be not totally convincing, earning one mark out of two. A few candidates fell short of full credit in part (c) by working on $(z^*)^2$ but not mentioning $-z^2$.

Q10.

The great majority of candidates showed that they had the necessary knowledge of complex numbers to cope with this very straightforward question. In a distressingly high number of instances the work was marred by elementary errors in the algebra, most commonly by a sign error causing $-z^*$ to appear as $-x - iy$. Many candidates also failed to indicate clearly in part (a) which were the real and imaginary parts, though many recovered the mark by using the real and imaginary parts correctly in part (b) of the question.

Q11.

Almost every candidate gave the right answer to part (a), and many went on to use the relationships between the roots and coefficients of a quadratic equation to answer part (b) correctly, though quite a number did not insert a minus sign for the coefficient of x . Some candidates followed other approaches, such as substituting the known root into the equation, expanding and equating real and imaginary parts, which if done carefully usually led to the right answers.

Q12.

In part (a) (i) of this question most candidates showed some knowledge of complex numbers but failed to display clearly the real and imaginary parts asked for in the question. Part (a) (ii) appeared to cause candidates no trouble at all; but by way of

contrast, very few candidates saw what was required in part (b): many equated the real part to zero, while others equated the real and imaginary parts to each other.

Q13.

The great majority of candidates found this question a good starter and obtained full marks without too much trouble. Careless errors sometimes caused candidates to miss out on one of the method marks for the question.

Q14.

Here again most candidates answered confidently and accurately. In part (a) many failed to state clearly which was the real part and which the imaginary part, though they recovered ground by using the correct expressions in part (b). As already stated, the solution of the simultaneous equations was often attempted by a substitution method.

Whichever method was used, numerical and sign errors were fairly common, but most candidates obtained the correct values for x and y . Two faults which were condoned this time were, in part (a), the retention of the factor 'i' in the imaginary part, and, in part (b) following correct values of x and y , a failure to give the final value of z correctly.

Q15.

Although most of the candidates who took this paper are good at algebra, there are still quite a number who made elementary sign errors. The fourth term of the expansion in part (a) frequently came out with a plus instead of a minus. The error of including an 'i' in the imaginary part was condoned.

In part (b) some candidates seemed not to realise that they needed to equate real and imaginary parts, despite the hint in part (a). Others used 16 for both the real and imaginary parts of the right-hand side of the equation. But a large number solved correctly to obtain full marks. Those who had made the sign error already referred to were given full marks in part (b) if their work was otherwise faultless.

Q16.

Part (a)(i) was answered well by the majority of candidates, though the fourth term in the expansion often appeared as $-i^2$ instead of $-i^2 \sqrt{5}$, and $-i^2$ was frequently simplified to -1 in the next line.

The next part, (a)(ii), tested the candidates' knowledge of technique to some extent. Those who understood what they were being asked to do invariably realised that the necessary hard work had already been done in part (a)(i), so that very little writing was needed to pick up two further marks. Those who thought they had to prove, rather than verify, the given statement were awarded one mark for showing a knowledge of the conjugate of z , but then spent much valuable time trying to solve the equation. There were three indications that this was not what the question required: the word 'Hence', the word 'verify', and the award of only two marks.

Part (b) of this question was generally well answered, though some candidates did not seem to be aware of the implications of the statement that the coefficients in the quadratic equation were real. The answers to part (b)(ii) were sometimes given in terms of p and q , and in part (b)(iii) many candidates equated p to the sum of the roots instead of to this number preceded by a minus sign.

Q17.

Part (a) of this question seemed to take many candidates unawares. They knew that the conjugate of z was $x - iy$, but claimed that the conjugate of $z + i$ was $x - iy + i$ or that it was simply $z - i$. Most candidates were aware of the required approach to part (b). They expanded the right-hand side of the equation and then equated the real and imaginary parts, giving two simple simultaneous equations which they could solve. In many cases their efforts were marred by sign errors or by the omission of a 2 when expanding the term $2iz$.

Q18.

Many candidates showed a sufficient familiarity with complex numbers to pick up a fair number of marks in this question. Part (a) was almost always correct, but a sign error spoiled many answers to part (b). Many candidates did not clearly state the real and imaginary parts in part (b), or stated the imaginary part as the correct answer multiplied by i , but these faults were condoned if, as usually happened, the candidate used the real and imaginary parts correctly in part (c).